

Google - 100 Technical MCQ Questions

Google (Software Engineer, New Grad)

For BTech Computer Science / IT Students

Object-Oriented Programming

1. What is a 'friend' function in C++?

- A) A function that has access to private/protected members of a class
- B) A public function only
- C) A function that cannot access class members
- D) A default constructor

2. Which of the following is true regarding static methods and polymorphism?

- A) Static methods support runtime polymorphism
- B) Static methods are resolved at compile time and do not support runtime polymorphism
- C) Static methods cannot be called
- D) Static methods must be abstract

3. Which of these is a valid example of encapsulation in practice?

- A) Making instance variables private and exposing getters/setters
- B) Making all variables public
- C) Using global variables everywhere
- D) Avoiding classes altogether

4. What is the diamond problem in OOP?

- A) An ambiguity arising from multiple inheritance of the same base class
- B) A memory leak issue
- C) A sorting algorithm's worst case
- D) A type of exception

5. Which of the following correctly describes 'late binding'?

- A) Method call resolved at compile time
- B) Method call resolved at runtime based on object type
- C) A binding used only for static variables
- D) A binding used for interfaces only

6. Which design principle states 'favor composition over inheritance'?

- A) SOLID-related OOP design guideline
- B) A rule that disallows interfaces
- C) A Java-only rule
- D) A database design rule

7. Which of these is true about interfaces in Java 8+?

- A) They can have default and static methods with bodies
- B) They can have instance variables with values changed at runtime
- C) They support constructors
- D) They cannot be implemented by multiple classes

8. What is operator overloading?

- A) Giving additional meaning to an operator for user-defined types
- B) Removing operators from a language
- C) A compilation error
- D) A runtime exception

9. In Python, which method is automatically called when an object is created?

- A) `__new__` then `__init__`
- B) `__del__`
- C) `__call__`
- D) `__str__`

10. What is object composition?

- A) Building complex objects by combining simpler objects
- B) Inheriting from multiple classes
- C) Creating static objects
- D) Deleting objects

DBMS & SQL

11. What does the term 'referential integrity' ensure?

- A) That foreign key values always match a primary key value in the referenced table (or are NULL)
- B) That all columns are indexed
- C) That all queries run faster
- D) That tables cannot be dropped

12. Which of the following best defines a stored procedure?

- A) A precompiled collection of SQL statements stored in the database and executed as a unit
- B) A type of foreign key
- C) A column constraint
- D) A backup mechanism

13. What is the difference between WHERE and HAVING clauses?

- A) WHERE filters rows before grouping, HAVING filters groups after aggregation
- B) They are identical in function
- C) HAVING is used only with SELECT *
- D) WHERE cannot use comparison operators

14. Which SQL keyword is used to remove duplicate rows from a result set?

- A) UNIQUE
- B) DISTINCT
- C) FILTER
- D) ONLY

15. What is a self-join?

- A) A join of a table with itself
- B) A join between two different databases
- C) A join without any condition
- D) An invalid SQL operation

16. Which normal form eliminates transitive dependency?

- A) 1NF
- B) 2NF
- C) 3NF
- D) 0NF

Operating Systems

17. What is a deadlock?

- A) A state where two or more processes are waiting indefinitely for resources held by each other
- B) A process that has finished execution
- C) A type of memory leak
- D) A CPU scheduling technique

18. Which of the following best defines 'mutual exclusion' as a deadlock condition?

- A) A resource can be used by multiple processes simultaneously
- B) At least one resource must be held in a non-shareable mode
- C) Processes never request resources
- D) Resources are always preemptable

19. What is the difference between a hard link and a symbolic link in a file system?

- A) A hard link points directly to the inode; a symbolic link is a separate file containing a path to the target
- B) They are identical
- C) Symbolic links cannot be broken
- D) Hard links can span across different file systems while symbolic links cannot

20. What is an inode in a Unix-like file system?

- A) A data structure storing metadata about a file (permissions, size, pointers to data blocks) but not its name
- B) The file's name only
- C) A type of directory
- D) A kernel module

21. Which scheduling algorithm can cause starvation?

- A) Round Robin
- B) Priority Scheduling (without aging)
- C) FCFS
- D) None of these

22. What is the primary function of an operating system?

- A) To compile programs
- B) To manage hardware resources and provide services to applications
- C) To design websites
- D) To encrypt files only

23. What is the role of the dispatcher in CPU scheduling?

- A) To select which process runs next based purely on priority
- B) To give control of the CPU to the process selected by the short-term scheduler
- C) To manage disk I/O only
- D) To handle network requests

24. Which file system attribute determines the smallest unit of disk space allocation?

- A) Sector/Block size
- B) File descriptor
- C) Inode number
- D) Directory depth

Computer Networks

25. Which layer of the OSI model is responsible for routing?

- A) Data Link Layer
- B) Network Layer
- C) Transport Layer
- D) Session Layer

26. What is the default port number for HTTP?

- A) 21
- B) 25
- C) 80
- D) 443

27. What is the main difference between TCP and UDP?

- A) TCP is connectionless, UDP is connection-oriented
- B) TCP is connection-oriented and reliable, UDP is connectionless and faster but unreliable
- C) There is no difference
- D) UDP guarantees ordered delivery while TCP does not

28. Which protocol is used to resolve a domain name to an IP address?

- A) FTP
- B) DNS
- C) SMTP
- D) ARP

29. What is the default port number for HTTPS?

- A) 80
- B) 21
- C) 443
- D) 22

30. Which protocol operates at the Transport layer and is connection-oriented?

- A) UDP
- B) IP
- C) TCP
- D) ICMP

Data Structures & Algorithms

- 31. What is the time complexity of building a heap from an unsorted array (heapify)?**
- A) $O(n \log n)$
 - B) $O(n)$
 - C) $O(\log n)$
 - D) $O(n^2)$
- 32. Which of the following problems is a classic example solved using dynamic programming?**
- A) Binary search
 - B) 0/1 Knapsack problem
 - C) Bubble sort
 - D) Linear search
- 33. Which data structure uses FIFO (First In First Out) order?**
- A) Stack
 - B) Queue
 - C) Tree
 - D) Graph
- 34. What is a balanced binary search tree used to prevent?**
- A) Overflow errors
 - B) Degeneration into a linked list (worst-case $O(n)$ operations)
 - C) Memory leaks
 - D) Integer overflow
- 35. What is the key property of a min-heap?**
- A) The parent node is always greater than its children
 - B) The parent node is always smaller than or equal to its children
 - C) It must be a complete binary tree only, with no ordering
 - D) It is always a balanced BST
- 36. Which of these best describes the 'sliding window' technique?**
- A) Maintaining a subset of elements (a window) over a sequence and adjusting its bounds to solve subarray/substring problems efficiently
 - B) A sorting technique for linked lists
 - C) A hashing technique
 - D) A tree traversal method
- 37. Which data structure is best suited for implementing a priority queue efficiently?**
- A) Array
 - B) Linked List
 - C) Heap
 - D) Stack
- 38. Which algorithm is used to find the Minimum Spanning Tree of a graph?**
- A) Dijkstra's Algorithm
 - B) Kruskal's or Prim's Algorithm
 - C) Bellman-Ford Algorithm
 - D) Floyd-Warshall Algorithm

39. Which sorting algorithm is considered stable and has $O(n \log n)$ time complexity in all cases?

- A) QuickSort
- B) Merge Sort
- C) Selection Sort
- D) Heap Sort

40. What is a monotonic stack typically used for?

- A) Finding the next greater/smaller element efficiently
- B) Sorting an array in place
- C) Implementing recursion
- D) Hashing

41. Which of the following is a common technique to resolve hash collisions?

- A) Chaining or open addressing
- B) Binary search
- C) Recursion
- D) Memoization

42. What is the average time complexity of searching in a well-implemented hash table?

- A) $O(n)$
- B) $O(\log n)$
- C) $O(1)$
- D) $O(n^2)$

43. What is the time complexity of inserting an element at the head of a singly linked list?

- A) $O(n)$
- B) $O(1)$
- C) $O(\log n)$
- D) $O(n \log n)$

44. Which data structure underlies the implementation of recursion internally?

- A) Queue
- B) Stack (call stack)
- C) Heap (memory heap)
- D) Graph

45. What is the time complexity of accessing an element in an array by index?

- A) $O(n)$
- B) $O(\log n)$
- C) $O(1)$
- D) $O(n \log n)$

46. Which data structure uses LIFO (Last In First Out) order?

- A) Queue
- B) Stack
- C) Linked List
- D) Array

47. What is dynamic programming primarily used for?

- A) Solving problems by breaking them into overlapping subproblems and storing results to avoid recomputation
- B) Sorting arrays only
- C) Random number generation
- D) Compiling code faster

48. What is a hash collision?

- A) When two different keys map to the same hash bucket/index
- B) When a hash table runs out of memory
- C) When a key is deleted incorrectly
- D) When two hash tables merge

49. What is the time complexity of finding the minimum element in an unsorted array?

- A) $O(1)$
- B) $O(\log n)$
- C) $O(n)$
- D) $O(n \log n)$

50. What is the amortized time complexity of appending an element to a dynamic array (like Python's list or Java's ArrayList)?

- A) $O(n)$
- B) $O(\log n)$
- C) $O(1)$ amortized
- D) $O(n^2)$

51. Which data structure is typically used to implement BFS?

- A) Stack
- B) Queue
- C) Heap
- D) Array only

52. Which data structure would you use to check if a string of brackets is balanced?

- A) Queue
- B) Stack
- C) Array
- D) Linked List

53. Which technique explores all possible options and backtracks when a solution path fails, often used in puzzles like N-Queens?

- A) Greedy algorithm
- B) Backtracking
- C) Dynamic programming
- D) Binary search

54. What is the time complexity of Dijkstra's algorithm using a binary heap?

- A) $O(V^2)$
- B) $O(E \log V)$
- C) $O(V + E)$
- D) $O(E^2)$

- 55. Which data structure is typically used to implement DFS (iteratively)?**
- A) Queue
 - B) Stack
 - C) Heap
 - D) Hash table
- 56. What is the time complexity of binary search on a sorted array of n elements?**
- A) $O(n)$
 - B) $O(\log n)$
 - C) $O(n \log n)$
 - D) $O(1)$
- 57. What is the space complexity of an iterative approach to computing Fibonacci numbers with memoization storing all n results?**
- A) $O(1)$
 - B) $O(n)$
 - C) $O(n^2)$
 - D) $O(\log n)$
- 58. What is the time complexity of Floyd-Warshall algorithm for all-pairs shortest paths?**
- A) $O(V^2)$
 - B) $O(V^3)$
 - C) $O(E \log V)$
 - D) $O(V + E)$
- 59. What does BFS stand for in graph traversal?**
- A) Best First Search
 - B) Breadth First Search
 - C) Binary First Search
 - D) Backtrack First Search
- 60. What is the primary difference between a greedy algorithm and dynamic programming?**
- A) Greedy makes a locally optimal choice at each step without reconsidering, DP considers overlapping subproblems and often guarantees a globally optimal solution
 - B) They are the same technique
 - C) DP never uses recursion
 - D) Greedy always gives an optimal solution while DP does not
- 61. Which of the following is true about a doubly linked list compared to a singly linked list?**
- A) It uses less memory per node
 - B) Each node has pointers to both the next and previous nodes, allowing traversal in both directions
 - C) It cannot be traversed backward
 - D) It has no advantages
- 62. What is the time complexity of the Two-Pointer technique on a sorted array to find a pair with a given sum?**
- A) $O(n^2)$
 - B) $O(n \log n)$
 - C) $O(n)$
 - D) $O(\log n)$

63. What is the worst-case time complexity of QuickSort?

- A) $O(n \log n)$
- B) $O(n^2)$
- C) $O(\log n)$
- D) $O(n)$

64. What is the time complexity of Bubble Sort in the worst case?

- A) $O(n)$
- B) $O(n \log n)$
- C) $O(n^2)$
- D) $O(\log n)$

65. Which traversal of a Binary Search Tree gives elements in sorted order?

- A) Pre-order
- B) Post-order
- C) In-order
- D) Level-order

66. What is memoization?

- A) Storing the results of expensive function calls and reusing them when the same inputs occur again
- B) A sorting technique
- C) A type of recursion without a base case
- D) A method to delete duplicate array elements

67. Which algorithmic technique does Merge Sort primarily use?

- A) Greedy
- B) Divide and Conquer
- C) Dynamic Programming
- D) Backtracking

68. What is tail recursion?

- A) A recursive call that is the last operation in a function, which can be optimized by some compilers into iteration
- B) A recursive function with no base case
- C) A function that calls itself twice
- D) A type of infinite loop

69. Which of the following best describes a trie data structure?

- A) A tree-like structure used for efficient retrieval of strings, especially for prefix matching
- B) A type of balanced binary search tree
- C) A hashing algorithm
- D) A graph traversal method

70. Which graph algorithm is used to find the shortest path in a graph with possible negative edge weights (no negative cycles)?

- A) Dijkstra's Algorithm
- B) Bellman-Ford Algorithm
- C) Prim's Algorithm
- D) Kruskal's Algorithm

Programming Language Fundamentals

71. What does 'pass by reference' mean?

- A) A copy of the variable's value is passed to the function
- B) The function receives a reference to the actual variable, so changes affect the original
- C) Variables cannot be passed to functions
- D) Only primitive types can be passed

72. What is the purpose of the 'finally' block in exception handling (Java/Python)?

- A) It executes only if an exception occurs
- B) It executes only if no exception occurs
- C) It always executes regardless of whether an exception occurred or not
- D) It is used to declare exceptions

73. In Python, which of the following is a mutable data type?

- A) Tuple
- B) String
- C) List
- D) int

74. In Java, what is the difference between '==' and '.equals()' when comparing objects?

- A) They always behave identically
- B) '==' compares references while '.equals()' compares content (if overridden)
- C) '.equals()' is only for primitives
- D) '==' is only used for Strings

75. What is the difference between a compiler and an interpreter?

- A) A compiler translates the whole program to machine code before execution, an interpreter executes code line-by-line
- B) They are identical
- C) Interpreters are always faster than compilers
- D) Compilers cannot detect syntax errors

76. Which of the following best describes garbage collection?

- A) A manual process where the developer frees memory explicitly
- B) An automatic process that reclaims memory occupied by objects no longer in use
- C) A compilation step
- D) A type of exception handling

77. Which keyword is used to handle exceptions in Python?

- A) catch
- B) except
- C) rescue
- D) handle

78. What does 'immutable' mean for an object (e.g., a String in Java)?

- A) Its value can be changed after creation
- B) Its state cannot be changed after it is created
- C) It can only be used once
- D) It cannot be passed to functions

79. What is a lambda function/expression?

- A) A named function stored in a file
- B) An anonymous, inline function typically used for short operations
- C) A type of class
- D) A compiler directive

80. What is the default value of an uninitialized boolean instance variable in Java?

- A) true
- B) false
- C) null
- D) 0

81. Which of the following is a value type (not a reference type) in Java?

- A) int
- B) String
- C) Array
- D) Object

82. In C, what does the 'malloc' function do?

- A) Deallocates memory
- B) Allocates a block of memory dynamically
- C) Initializes a variable to zero
- D) Declares a static variable

83. What is a memory leak?

- A) When allocated memory is never freed even though it is no longer needed, wasting resources over time
- B) When a program uses too little memory
- C) A type of compilation error
- D) A CPU scheduling issue

Web & Cloud Fundamentals

84. What is a RESTful API?

- A) An API that strictly requires SOAP protocol
- B) An architectural style for designing networked applications using stateless operations over standard HTTP methods
- C) A type of database
- D) A programming language

85. Which HTTP method is typically used to retrieve data without side effects?

- A) POST
- B) GET
- C) DELETE
- D) PUT

86. What is the primary difference between HTTP and HTTPS?

- A) HTTPS uses encryption (SSL/TLS) to secure data in transit while HTTP does not
- B) HTTP is faster in all cases
- C) HTTPS cannot use port 443
- D) There is no difference

87. What does CSS stand for?

- A) Cascading Style Sheets
- B) Computer Style Sheets
- C) Creative Style System
- D) Cascading Software Sheets

88. What does HTTP stand for?

- A) HyperText Transfer Protocol
- B) HighText Transmission Protocol
- C) Hyperlink Transfer Text Protocol
- D) HyperText Trade Protocol

89. What does 'statelessness' mean in the context of REST APIs?

- A) The server stores client session state between requests
- B) Each request from a client contains all the information needed to process it, with no reliance on stored server-side session state
- C) The API has no state at all, including no data
- D) Stateless APIs cannot use authentication

90. What is the purpose of cookies in web applications?

- A) To store small pieces of data on the client side to maintain state/session information
- B) To encrypt HTTPS traffic
- C) To speed up DNS resolution
- D) To compress images

System Design Basics

91. What is idempotency in the context of API design?

- A) An operation that produces the same result no matter how many times it is performed
- B) An operation that can never be repeated
- C) An operation that always fails on retry
- D) A synonym for authentication

92. What is vertical scaling?

- A) Adding more machines to a system
- B) Increasing the resources (CPU/RAM) of an existing machine
- C) Distributing data across regions
- D) A type of caching

93. What is eventual consistency?

- A) A guarantee that all replicas are updated instantly
- B) A consistency model where, given enough time without new updates, all replicas converge to the same value
- C) A model where data is never consistent
- D) A synonym for strong consistency

94. What is caching used for in system design?

- A) To permanently store all application data
- B) To temporarily store frequently accessed data closer to the application for faster retrieval
- C) To replace a database entirely
- D) To handle user authentication

95. What is the CAP theorem in distributed systems?

- A) A system can simultaneously guarantee Consistency, Availability, and Partition tolerance
- B) A distributed system can only guarantee two of Consistency, Availability, and Partition tolerance at the same time during a network partition
- C) A theorem about compiler design
- D) A theorem unrelated to distributed systems

96. What is a load balancer used for in system design?

- A) To store data persistently
- B) To distribute incoming requests across multiple servers to avoid overload on any single server
- C) To compress images
- D) To manage user authentication only

97. What is the purpose of an API Gateway in a microservices architecture?

- A) To act as a single entry point that routes requests to appropriate backend services and can handle cross-cutting concerns like auth and rate limiting
- B) To store all application data
- C) To replace load balancers entirely
- D) To compile microservices code

98. What is the purpose of a message queue (e.g., Kafka, RabbitMQ) in system design?

- A) To decouple producers and consumers of data, enabling asynchronous processing and buffering
- B) To replace a database
- C) To render a webpage
- D) To manage user sessions only

99. What is horizontal scaling?

- A) Adding more power (CPU/RAM) to an existing server
- B) Adding more servers/machines to distribute the load
- C) Reducing the number of servers
- D) Only applicable to databases

100. What is database sharding used for?

- A) To horizontally partition a large database across multiple servers to improve scalability
- B) To encrypt a database
- C) To back up data
- D) To normalize tables

Answer Key

1-A	2-B	3-A	4-A	5-B	6-A	7-A	8-A	9-A	10-A
11-A	12-A	13-A	14-B	15-A	16-C	17-A	18-B	19-A	20-A
21-B	22-B	23-B	24-A	25-B	26-C	27-B	28-B	29-C	30-C
31-B	32-B	33-B	34-B	35-B	36-A	37-C	38-B	39-B	40-A
41-A	42-C	43-B	44-B	45-C	46-B	47-A	48-A	49-C	50-C
51-B	52-B	53-B	54-B	55-B	56-B	57-B	58-B	59-B	60-A
61-B	62-C	63-B	64-C	65-C	66-A	67-B	68-A	69-A	70-B
71-B	72-C	73-C	74-B	75-A	76-B	77-B	78-B	79-B	80-B
81-A	82-B	83-A	84-B	85-B	86-A	87-A	88-A	89-B	90-A
91-A	92-B	93-B	94-B	95-B	96-B	97-A	98-A	99-B	100-A

Format: [Question No.]-[Correct Option Letter]. Content compiled from publicly available 2026 placement-prep resources and established CS-fundamentals question banks - verify current-year specifics before your interview.